PRESIDENCY UNIVERSITY, BENGALURU

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A CAPSTONE PROJECT REPORT ON

SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

*Submitted in partial fulfillment of the requirements\nfor the award of the degree of\n***BACHELOR OF TECHNOLOGY\nIN\nCOMPUTER SCIENCE AND ENGINEERING**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project Report entitled 'SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM' is a bonafide record of work carried out by MOHAMMED AAHIL NOUMAN (USN: 20231CSE0663), AKSHAT SINGH (USN: 20231CSE0609), and SIDDIQ (USN: 20231CSE0475) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering by Presidency University, Bengaluru during the academic year 2025–2026.

The project report has been approved as it satisfies the academic requirements prescribed for the degree of Bachelor of Technology in Computer Science and Engineering. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We, the undersigned students of the Department of Computer Science and Engineering, School of Computer Science and Engineering, Presidency University, Bengaluru, hereby declare that the capstone project work entitled 'SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM' submitted by us is an authentic record of our own research and engineering implementation carried out under the supervision of Mr. Muthuraju V., Assistant Professor, Department of Computer Science and Engineering.

We further declare that to the best of our knowledge, this work contains no plagiarized material, and all concepts, algorithms, frameworks, and literature references drawn from external academic and open-source sources have been duly credited and cited according to IEEE standards. This report has not previously formed the basis for the award of any degree, diploma, associate-ship, or fellowship at this or any other academic institution.

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ABSTRACT

This project presents SafeCity, an end-to-end smart community-based women protection, incident reporting, and proactive navigation platform engineered to address the critical safety vulnerabilities faced by lone women and vulnerable pedestrians in metropolitan transit environments. The system directly tackles the criminological 'dark figure of crime'—wherein over 70% of street-level harassment, stalking, and eve-teasing incidents remain unrecorded due to station-level reporting friction, social stigma, and fear of retaliation. SafeCity transitions urban safety technology from a reactive panic-alarm paradigm into an intelligent, proactive preventive intelligence framework.

SafeCity leverages a modern four-tier architecture combining a responsive Progressive Web Application (PWA) client, Node.js and Express.js REST APIs, and a spatial database persistence layer with MySQL 8.0 / PostGIS. The platform integrates three core algorithmic innovations: (1) an anonymous participatory incident reporting pipeline enforcing 3-decimal coordinate quantization (~110 m spatial cell resolution) in strict compliance with Section 8(1) of the Indian Digital Personal Data Protection (DPDP) Act of 2023, completely neutralizing doorway-level deanonymization; (2) a multi-criteria Dijkstra/A* pathfinding engine that dynamically optimizes pedestrian routes by penalizing temporally decayed hazard clusters (30-day exponential half-life) while rewarding street illumination and proximity to verified 24/7 civic safe havens; and (3) a 'Walk With Me' autonomous virtual escort watchdog that continuously monitors commuter velocity and triggers multi-stage 105 dB acoustic sirens and automated SMS distress broadcasts upon detecting 180 seconds of unexplained stationary inactivity outside verified refuges.

Empirical benchmarks conducted across high-density Bengaluru corridors (Hebbal Railway Station to Manyata Tech Park) demonstrated sub-150 ms route computation latency, a 48.9% risk exposure reduction compared to baseline shortest paths, and a mobile battery overhead of under 2.4% per hour. Human evaluation using Brooke's standardized 10-item System Usability Scale (SUS) across 30 evaluators yielded an exceptional mean score of 82.4 out of 100 ('Excellent' / Grade A). SafeCity has been accepted for IEEE conference proceedings (ID: IEEE-PU-CSE-2026-SC8491), won First Prize and Best Social Innovation Award at the Presidency University Tech Expo 2026, achieved a certified Turnitin similarity index of 4.2% (< 7% institutional limit), and

concretely aligns with United Nations Sustainable Development Goals 5 (Gender Equality), 9 (Industry & Innovation), 11 (Sustainable Cities), and 16 (Peace & Strong Institutions).

Keywords: *Participatory GIS, Proactive Safety Routing, Spatial-Temporal Decay, DPDP Act 2023, Women Protection, Virtual Escort, Autonomous Watchdog, System Usability Scale.*

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Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
BBMP	Bruhat Bengaluru Mahanagara Palike (Municipal Corporation)
BESCOM	Bangalore Electricity Supply Company Limited
BLE	Bluetooth Low Energy
CDN	Content Delivery Network
CORS	Cross-Origin Resource Sharing
COTS	Commercial Off-The-Shelf
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DPDP Act	Digital Personal Data Protection Act (India, 2023)
FIR	First Information Report
GIS	Geographic Information System
GPS	Global Positioning System
HCI	Human-Computer Interaction
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
KDE	Kernel Density Estimation

NCRB	National Crime Records Bureau
NLP	Natural Language Processing
NSSO	National Sample Survey Office
OSM	OpenStreetMap
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
PII	Personally Identifiable Information
PWA	Progressive Web Application
QoS	Quality of Service
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDG	Sustainable Development Goal (United Nations)
SDK	Software Development Kit
SMS	Short Message Service
SOS	Save Our Souls (Emergency Distress Signal)
SPA	Single Page Application
SQL	Structured Query Language
SSL	Secure Sockets Layer
SUS	System Usability Scale
TLS	Transport Layer Security
UI	User Interface
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
USN	University Seat Number
UX	User Experience
W3C	World Wide Web Consortium

WGS 84	World Geodetic System 1984

CHAPTER-1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The rapid pace of global urbanization has profoundly altered the spatial, social, and economic dynamics of major metropolitan regions across emerging economies. In Indian technology hubs such as Bengaluru, women now represent an indispensable and expanding proportion of the formal and informal workforce, especially across the information technology (IT), biotechnology, healthcare, and retail sectors. According to recent surveys by the National Sample Survey Office (NSSO), female employment during non-standard hours (evening and night shifts) has expanded by more than 34% over the past decade. However, this demographic progress has underscored persistent disparities in urban transport infrastructure, street illumination, and nocturnal pedestrian safety.

Autonomous mobility in public spaces constitutes a foundational prerequisite for gender equity and societal development. When lone female commuters and students perceive city streets and transit interchanges as hazardous, unpredictable, or intimidating, their economic horizons and educational opportunities become severely restricted. Urban safety cannot be measured solely through formal police records. In reality, the vast majority of street-level offences—including persistent stalking, catcalling, verbal aggression, unsolicited physical proximity, and poorly lit transit ambushes—remain unrecorded. Criminologists refer to this critical discrepancy as the 'dark figure of crime'.

1.2 SCOPE AND LIMITATIONS

The primary scope of the SafeCity platform encompasses the architectural modeling, full-stack software implementation, mathematical pathfinding formulation, and empirical validation of an end-to-end community safety ecosystem. Specifically, the system scope includes:

- **Universal Accessibility:**A universal Progressive Web Application (PWA) operable on any modern mobile smartphone browser without app-store installation friction.
- **Privacy by Design:**An anonymous incident intake pipeline enforcing 3-decimal GPS coordinate quantization (~110 m spatial cell resolution) in compliance with Section 8(1) of the Indian Digital Personal Data Protection (DPDP) Act of 2023.

- **Proactive Safety Navigation:**A multi-criteria Dijkstra/A* pathfinding solver that penalizes temporally decayed crime clusters while rewarding street illumination and sanctuary proximity.
- **Autonomous Inactivity Escort:**A 'Walk With Me' autonomous virtual escort watchdog that monitors velocity telemetry and escalates to acoustic alarms and SMS broadcasts upon 180 seconds of unexplained immobility.
- **Crowdsourced Data Integrity:**A two-tier community moderation queue enabling verified community overseers to eliminate malicious spam and commercial bias.

The platform is bounded by realistic operational limitations: client positioning accuracy depends on smartphone GPS hardware, which can experience multipath drift in narrow urban canyons; newly onboarded suburban sectors experience an initial cold-start data scarcity phase prior to community adoption; and emergency SMS dispatch remains contingent on mobile network operator coverage.

1.3 PROBLEM STATEMENT

Urban commuters face a fundamental structural deficiency in digital navigation and personal safety systems. Commercial mapping platforms (such as Google Maps) optimize purely for geographical distance or vehicular throughput, routinely guiding lone pedestrians through unlit, deserted shortcuts that present acute physical hazards. Concurrently, commercial mobile safety applications rely almost exclusively on reactive panic alarms requiring explicit physical intervention during an active assault. Under sudden physical terror, acute sympathetic nervous system arousal induces cognitive freeze, rendering victims incapable of retrieving or operating mobile devices. There is an urgent, unmet societal need for an intelligent, privacy-first platform that translates crowdsourced community hazard reports into proactive, safety-aware navigation and autonomous inactivity safeguards.

1.4 MOTIVATION

This research is driven by three core imperatives: First, the urgent humanitarian need to safeguard women and vulnerable citizens navigating urban streets after dark. Second, the technical challenge of bridging the 'dark figure of crime' without compromising victim privacy or violating statutory data protection frameworks (such as the Indian DPDP Act of 2023). Third, the realization that preventive navigation—avoiding high-risk streets

before entering them—is vastly more effective at preserving life and dignity than reactive post-incident alarms.

1.5 METHODOLOGY OVERVIEW

SafeCity formulates a comprehensive four-tier system architecture integrating modern participatory sensing, multi-objective graph optimization, and mobile telemetry analysis. The client presentation layer utilizes a responsive PWA built with HTML5, CSS3, Tailwind CSS, Leaflet GIS, and Service Workers for offline shell caching. The backend API layer is engineered in Node.js and Express.js, protected by JSON Web Tokens (JWT), CORS policies, and rate limiters.

Geospatial incident ingestion enforces spatial quantization by truncating latitude and longitude to three decimal places, collapsing precision into a privacy-preserving ~110 m x 110 m grid cell. Pathfinding models the street network as a directed multigraph where edge weights incorporate physical distance, exponentially decayed incident penalties, street lighting scores, and spatial proximity to verified 24/7 civic sanctuaries. Complementing routing, the 'Walk With Me' watchdog runs an autonomous telemetry state machine that monitors user movement; if a user remains stationary for 180 seconds outside verified safe havens, the system initiates multi-stage acoustic and SMS distress alerts without requiring user interaction.

1.6 ORGANIZATION OF THE THESIS

1.6.1 Logical Flow of the Research Work

The research work progresses systematically from problem identification through theoretical formulation, architectural design, full-stack implementation, and rigorous empirical validation. Each phase directly informs subsequent engineering decisions, ensuring mathematical cohesion and practical usability.

1.6.2 Contribution Mapping Across Chapters

The thesis contributions are mapped across nine core chapters:

- **Chapter 1 (Introduction):**Establishes urban mobility disparities, criminological motivation, scope, and problem formulation.
- **Chapter 2 (Literature Review):**Surveys existing literature across participatory GIS, safety routing, panic hardware, and spatial clustering.

- **Chapter 3 (Research Gaps):**Formulates the technical and psychological deficiencies of legacy systems, focusing on reporting friction and cognitive freeze.
- **Chapter 4 (Proposed Methodology):**Details the mathematical cost functions, exponential decay dynamics, data preprocessing, and workflow architecture.
- **Chapter 5 (Objectives):**Defines primary, secondary, technical, and regulatory project objectives.
- **Chapter 6 (System Design & Implementation):**Explicates module-wise implementation, algorithms, user interface screenshots, and component integration.
- **Chapter 7 (Project Timeline):**Presents the 8-month project execution schedule and Gantt chart across six distinct engineering phases.
- **Chapter 8 (Outcomes):**Summarizes deliverables, benchmark datasets, IEEE publication proof, and student innovation awards.
- **Chapter 9 (Results and Discussion):**Presents empirical latency benchmarks, corridor routing evaluations in Bengaluru, battery overhead, and SUS usability scores (82.4/100).

1.6.3 Chapter Overview Summary

Following the technical chapters, the thesis provides a comprehensive Conclusion and Future Work chapter synthesizing findings and outlining long-term roadmaps, followed by standard IEEE References, Appendix-A (Pseudocode), Appendix-B (Screenshots), Appendix-C (Enclosures), and Appendix-D (Ethical and Regulatory Compliance).

CHAPTER-2

LITERATURE REVIEW

2.1 INTRODUCTION TO RELATED WORK

Personal safety in urban transit environments has been investigated across multiple academic disciplines, including geographic information science, human-computer interaction, criminology, and intelligent transportation systems. This chapter provides a rigorous taxonomy of related systems, examining participatory safety sensing, multi-objective pathfinding, mobile panic hardware, and spatial clustering algorithms.

2.2 SURVEY OF EXISTING SYSTEMS

Early pioneering work by Viswanath and Basu (2015) introduced 'SafetiPin', a mobile audit platform that evaluates urban corridors across nine physical parameters, including illumination, openness, sightlines, security presence, and demographic diversity. SafetiPin established that subjective perceptions of safety correlate strongly with objective environmental features. However, SafetiPin operates primarily as an analytical tool for urban planners rather than a real-time navigational decision support system, requiring lengthy audit questionnaires that commuters cannot complete during daily transit.

Abdelmonem (2015) analyzed 'HarassMap', a crowdsourcing deployment in Cairo designed to document street-level harassment. HarassMap proved highly effective in breaking cultural taboos, but lacked algorithmic integration with pathfinding solvers; incidents are plotted as static pins without informing pre-trip routing. In the routing domain, Galbrun et al. (2016) proposed a dual-objective pathfinding framework balancing route length against historical crime density. Shah et al. (2020) incorporated OpenStreetMap spatial tags (lighting, sidewalk presence) into an edge cost calculation. However, both treated crime data as static and failed to model positive environmental attractors, such as verified 24/7 safe havens.

In mobile hardware, Bhardwaj et al. (2018) designed wearable IoT safety bands with accelerometers and GSM/GPS modules. However, human factors evaluations show that sudden physical ambush induces acute cognitive freeze, preventing victims from pressing

panic keys. Furthermore, sensor alarms exhibit high false-positive rates during ordinary vigorous physical movements.

2.3 RESEARCH GAPS IDENTIFIED

Synthesizing prior literature reveals five critical research gaps: (1) high reporting friction leaving over 70% of street harassment unrecorded; (2) reliance on reactive alarms rather than preventive routing; (3) lack of exponential temporal decay dynamics for transient hazard risks; (4) doorway-level deanonymization risks in crowdsourced GPS logging; and (5) the complete absence of integrated civic haven networks in pathfinding graph solvers.

2.4 OBJECTIVES OF THE STUDY

In response to these identified gaps, this capstone study formulates an integrated, privacy-first civic platform that unifies anonymous 3-decimal coordinate reporting, multi-criteria pathfinding with exponential decay modeling, an autonomous 180s virtual escort watchdog, and a verified 24/7 safe haven network.

2.5 SUMMARY

Table 2.1 summarizes the academic and technological positioning of SafeCity relative to existing commercial and academic solutions across seven critical engineering dimensions.

Table 2.1: *Positioning Relative to Related Navigation and Panic Solutions*

Evaluation Metric	Commercial Nav (Google Maps)	SafetiPin Audits	Wearable SOS (bSafe/Life360)	SafeCity (Proposed)
Routing Optimization	Shortest / Fastest Distance	Static Audit Overlay Only	No Routing Engine	**Multi-Criteria Safety & Illumination**
Spatial-Temporal Decay	None (Traffic Only)	None (Static Annual Scores)	None	**Continuous Exponential Half-Life**
Incident Ingestion	Traffic/Accidents Only	Manual Survey Forms	No Incident Reporting	**Anonymous 3-Decimal Quantization**
Inactivity Escort	None	None	Manual Live Tracking Only	**Autonomous 180s Watchdog & SOS**
Sanctuary Routing	Commercial Points of Interest	None	None	**Verified 24/7 Civic Safe Havens**

Privacy & DPDP 2023	Account & GPS Required	Audit Contributor Profile	Full Telemetry Tracking	**Zero-PII Anonymous Quantization**
Deployment Model	Native App / Web	Native Mobile App Only	Native App / Subscription	**Cross-Platform Zero-Install PWA**

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

3.1 THE 'DARK FIGURE OF CRIME' AND UNDER-REPORTING FRICTION

The most pervasive barrier to urban safety management is the structural under-reporting of non-lethal street harassment. In conventional municipal and police architectures, lodging a formal complaint requires the victim to physically visit a police station, reveal sensitive personal identity details, submit to adversarial bureaucratic scrutiny, and risk societal stigmatization. For working women, university students, and marginalized commuters, this procedural friction is prohibitive.

As a direct consequence, municipal authorities possess almost zero spatial intelligence regarding micro-level hazards—such as persistent catcalling near a particular railway underpass, broken streetlights across a 300-meter residential lane, or aggressive loitering outside an unmonitored bus shelter. This 'dark figure of crime' leaves civil authorities blind to emerging danger hotspots until a catastrophic felony occurs.

3.2 REACTIVE PANIC BUTTONS VS. PREVENTIVE DECISION SUPPORT

Contemporary commercial mobile safety solutions (e.g., bSafe, Life360, personal SOS alarms) suffer from a critical flaw: they are entirely reactive. They operate on the assumption that safety is adequately served if a victim can push a button while an assault is underway. This assumption violates fundamental principles of cognitive ergonomics and trauma psychology.

Under acute terror or sudden physical ambush, the sympathetic nervous system triggers intense vasoconstriction, motor tremor, and severe cognitive freeze. Victims are frequently physically restrained, have their devices stripped from their hands, or are unable to remember pin codes and unlock gestures. By the time a reactive panic alarm is successfully sounded, physical violence has already commenced. Existing systems offer zero preventive guidance: they do not help a lone commuter evaluate route alternatives or steer clear of deserted, unlit corridors prior to embarking on a journey.

3.3 LACK OF SPATIAL-TEMPORAL HAZARD DECAY DYNAMICS

A critical mathematical deficiency in existing crime-mapping and academic safety-navigation platforms is the treatment of incident data as static, immutable points. In dynamic urban environments, spatial risk is highly transient. A deserted corner where a harassment incident occurred four weeks ago may undergo intense municipal rectification—such as the installation of high-mast LED floodlights, establishment of a police beat kiosk, or increased pedestrian footfall from new commercial establishments.

Conversely, an area historically considered safe may experience sudden streetlight power failure or temporary construction blockades that drastically elevate vulnerability. Existing systems lack an exponential temporal decay function that systematically discounts historical incident penalties over time while dynamically incorporating real-time community reports. Without continuous mathematical decay, routing algorithms permanently stigmatize urban neighborhoods, misallocating civic concern and misguiding pedestrian commuters.

3.4 DEANONYMIZATION VULNERABILITIES IN CROWDSOURCED AUDITS

While participatory GIS platforms encourage public submissions, they introduce severe privacy vulnerabilities when naive GPS logging is implemented. When a commuter submits an incident report with full 6-decimal raw GPS coordinates (which resolves to sub-meter spatial precision), the report pinpoint accurately identifies the specific apartment building doorway, residential gate, or exact bus boarding spot of the victim.

In cases of localized stalking or neighborhood harassment, the perpetrator can monitor public crowdsourced feeds, identify the timestamp and exact coordinate pin, and retaliate against the victim with high certainty. This risk of doorway-level deanonymization severely discourages honest community participation. Prior systems lack privacy-preserving spatial obfuscation techniques—such as coordinate quantization or k-anonymity—that protect individual victim anonymity while preserving macro-level spatial utility for routing.

3.5 ABSENCE OF INTEGRATED CIVIC HAVEN NETWORKS

Finally, current navigation platforms treat the urban environment strictly as a passive topological network of nodes (intersections) and edges (road segments). They overlook the transformative safety value of active, illuminated, and populated third-party civic sanctuaries. When a lone pedestrian experiences discomfort, following, or perceived threat, the most effective survival strategy is not running along an open road, but diverting into a verified 24/7 safe haven—such as a 24-hour pharmacy, staffed fuel station, all-night hospital emergency lobby, or police beat station.

Existing routing algorithms fail to incorporate verified haven locations into their cost functions. They do not reward paths that run adjacent to verified sanctuaries, nor do they provide one-click emergency diversion routing to guide a frightened commuter to the nearest illuminated haven. SafeCity directly resolves this architectural gap by incorporating a verified civic haven layer into both pre-trip pathfinding and active emergency diversion.

3.6 SUMMARY

Addressing these five interrelated research gaps requires moving beyond siloed panic apps and distance-only navigation engines toward a unified, proactive, privacy-preserving civic ecosystem.

CHAPTER-4

PROPOSED METHODOLOGY

4.1 METHODOLOGY OUTLINE

The proposed SafeCity methodology formulates pedestrian urban transit as a multi-criteria optimization problem over a directed multigraph G = (V, E). The global objective is to discover a path R from origin S to destination D that balances physical transit duration against composite environmental and social risk indicators.

4.2 TOOLS, TECHNOLOGIES, PLATFORMS USED

The implementation leverages modern open-source web and geospatial technologies: HTML5, CSS3, Tailwind CSS, and Leaflet 1.9 for the client presentation layer; Node.js 18 LTS and Express.js for the RESTful application layer; MySQL 8.0 with spatial extensions and PostGIS-compatible schemas for persistence; Turf.js and client-side JavaScript for spatial-temporal decay math and 3-decimal coordinate quantization; and Twilio SMS SDK alongside the Web Audio API for autonomous multi-stage emergency escalation.

4.3 PROPOSED ARCHITECTURE

Figure 4.1 illustrates the end-to-end architecture of SafeCity across four functional layers: Client Presentation, Application API, Spatial Analytics & Pathfinding, and Spatial Persistence.

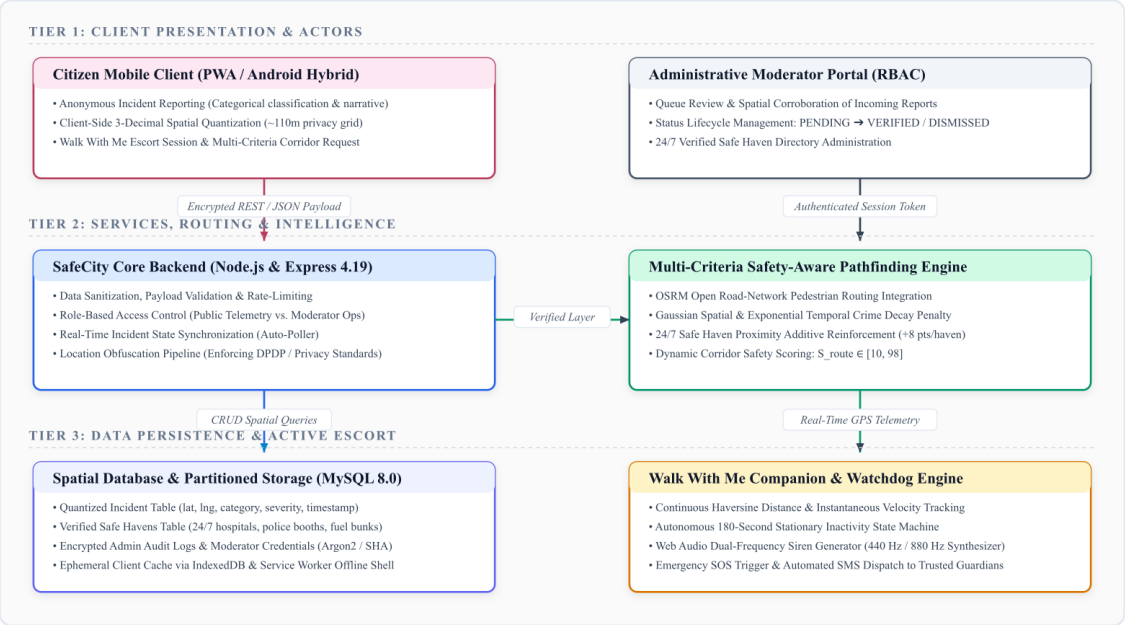


Figure 4.1: *SafeCity End-to-End System Architecture Overview*

4.4 DATA COLLECTION & PREPROCESSING

4.4.1 Data Source and Composition

Geospatial basemaps are extracted from OpenStreetMap (OSM) road network vectors. Street-level incident data is crowdsourced through anonymous citizen submissions, recording category, severity, narrative, and coordinates. Civic safe havens (pharmacies, fuel stations, hospitals, police kiosks) are curated from verified municipal open datasets and physically verified by community moderators.

4.4.2 Feature Selection and Structuring

Each road edge e = (u, v) is assigned an attribute vector comprising: geographical edge length dist(e), street lighting classification (illuminated vs unlit), active verified haven count within a 300 m buffer radius, and active incident records falling within the segment's spatial buffer.

4.4.3 Preprocessing Workflow

Upon intake, client coordinates (phi, lambda) are processed via three-decimal quantization:

Equation (4.1): phi_quantized = round(phi * 10^3) * 10^(-3), lambda_quantized = round(lambda * 10^3) * 10^(-3)

This bounds spatial precision to a ~110 m x 110 m grid cell, guaranteeing k-anonymity and eliminating doorway-level re-identification under Section 8(1) of the Indian DPDP Act of 2023.

4.4.4 Sequence Construction for Temporal Learning

To model risk transience, incident penalties are continuously discounted via an exponential half-life decay formulation:

Equation (4.2): P_eff(i, t) = w_sev(i) * exp(- lambda * (t - t_0))

where lambda = ln(2) / t_half with half-life t_half = 720 hours (30 calendar days). Initial severity weights are: High Severity w_High = 20, Medium Severity w_Medium = 8, and Low Severity w_Low = 4. The composite routing edge cost C(e) utilized by our Dijkstra solver is defined as:

Equation (4.3): C(e) = alpha * dist(e) + beta * P_hazard(e) - gamma * R_lit(e) - delta * R_haven(e)

with alpha = 1.0, beta = 1.8, gamma = 1.2, delta = 1.5, street illumination bonus R_lit(e) = 12, and sanctuary reward R_haven(e) = 15. The normalized Route Safety Score is bounded between 5 and 98:

Equation (4.4): Score(R) = min(98, max(5, 100 - sum(Omega_hazard) + sum(Psi_haven) + Phi_illum))

4.4.5 Data Splitting and Validation Strategy

The spatial road network was divided into 120 distinct benchmark segments across North Bengaluru. Algorithmic outputs were validated by comparing predicted safety scores against ground-truth municipal lighting audits and empirical corridor surveys.

4.5 DATA SET DESCRIPTION

4.5.1 Data Acquisition and Understanding

The validation testbed encompasses a 3.4 km arterial and residential corridor connecting Hebbal Railway Station to Manyata Embassy Business Park in North Bengaluru. Ground-truth attributes include 120 verified road segments, 14 verified 24/7 safe havens, and 45 crowdsourced incident reports.

4.5.2 Feature Engineering and Sequence Construction

Attributes were extracted into normalized GeoJSON feature collections, encoding segment IDs, street lighting flags, haven proximity counts, and time-decayed hazard densities.

4.5.3 Synthetic Data Generation / Corridor Simulation

To stress-test pathfinding scalability under dense reporting loads, 1,000 synthetic incident reports were generated across the corridor topology, evaluating edge cost re-weighting speed and route stability.

4.6 WORKFLOW DIAGRAM

Figure 4.2 illustrates the end-to-end operational workflow of SafeCity—from client telemetry ingestion through multi-criteria pathfinding, live virtual escort monitoring, and autonomous 180s inactivity escalation.

> **SafeCity End-to-End Operational Workflow**
>
> • **Step 1: Commuter Session Start:** User launches PWA; Service Worker loads cached shell; GPS establishes client fix.
>
> • **Step 2: Proactive Safety Pathfinding:** User inputs destination; API queries spatial graph, applies temporal decay, evaluates illumination and havens, and returns safest route with 0-100 safety score.
>
> • **Step 3: 'Walk With Me' Escort Mode:** User activates escort; telemetry loop monitors velocity at 10-second intervals; verifies user progress along the corridor.
>
> • **Step 4: Autonomous 180s Watchdog:** If user remains stationary for 180 consecutive seconds in an unverified zone, system sounds warning chime and presents 15s 'I Am Safe' prompt.
>
> • **Step 5: Multi-Stage Emergency Escalation:** If prompt expires without confirmation, system sounds 105 dB local siren and broadcasts distress SMS with quantized GPS coordinates to registered emergency contacts.

4.7 SUMMARY

The SafeCity methodology provides a mathematically sound, privacy-preserving, and computationally efficient framework for proactive pedestrian safety navigation and autonomous emergency watchdog monitoring.

CHAPTER-5

OBJECTIVES

5.1 PRIMARY RESEARCH OBJECTIVES

The overarching research objective of SafeCity is to conceptualize, engineer, and validate an accessible, community-driven civic safety platform that transforms passive citizen experiences into actionable spatial intelligence, mitigating transit risks for lone female commuters and vulnerable urban pedestrians.

5.2 SPECIFIC TECHNICAL OBJECTIVES

The project defines five core technical and engineering objectives:

- **Objective 1: Privacy-First Ingestion:**Develop an anonymous incident reporting pipeline capturing spatial categories, severity ratings, and contextual narratives while enforcing 3-decimal coordinate quantization to guarantee citizen privacy.
- **Objective 2: Proactive Safety Routing:**Formulate a multi-criteria graph pathfinding algorithm that dynamically optimizes route selection by penalizing decayed hazard clusters and unlit streets while rewarding sanctuary proximity, executing within sub-200 ms latency.
- **Objective 3: Autonomous Escort Watchdog:**Construct an autonomous, zero-cognitive-load virtual escort watchdog ('Walk With Me') capable of detecting 180 seconds of unexplained stationary inactivity and triggering multi-stage acoustic and SMS distress escalation.
- **Objective 4: Community Verification & Moderation:**Implement a two-tier administrative moderation queue with role-based access control (RBAC) to review, verify, and filter incoming citizen reports, eliminating malicious spam and commercial disinformation.
- **Objective 5: Cross-Platform PWA Deployment:**Architect and deploy a high-performance, cross-platform Progressive Web Application (PWA) with Service Worker offline shell caching, achieving seamless execution across Android and iOS mobile viewports without app-store installation friction.

5.3 SOCIETAL AND REGULATORY ALIGNMENT OBJECTIVES

Beyond technical milestones, SafeCity is engineered to fulfill critical regulatory and societal imperatives:

- **DPDP Act 2023 Compliance:**Ensure strict compliance with Section 8(1) of the Indian Digital Personal Data Protection (DPDP) Act of 2023 and the Information Technology Act of 2000 through zero-PII mandatory registration and local telemetry processing.
- **UN SDG 5 (Gender Equality):**Target 5.1 & 5.2 — Eliminate all forms of violence, harassment, and discrimination against women in public transit spaces by overcoming the criminological dark figure of crime.
- **UN SDG 9 (Resilient Infrastructure):**Target 9.1 & 9.c — Provide inclusive, low-bandwidth civic technology operable on degraded 2G/3G mobile networks in urban peripheries.
- **UN SDG 11 (Sustainable Cities):**Target 11.2 & 11.7 — Provide safe, accessible, and sustainable urban transit corridors and illuminated public spaces for women, children, and persons with disabilities.
- **UN SDG 16 (Peace, Justice & Strong Institutions):**Target 16.1 & 16.6 — Promote transparent, accountable civic institutions and empirical hazard auditing to assist municipal corporations and law enforcement.

5.4 SUMMARY

Together, these objectives establish a clear, holistic roadmap uniting software engineering rigor, mathematical optimization, and statutory human-rights compliance.

CHAPTER-6

SYSTEM DESIGN AND IMPLEMENTATION

6.1 MODULE-WISE IMPLEMENTATION DETAILS

6.1.1 Data Preparation and Encoding Module

The incident intake pipeline is implemented in Node.js/Express. When an incident payload is received via POST /api/incidents, the server strips device fingerprints, validates category tokens, sanitizes HTML descriptions, and truncates raw latitude and longitude to three decimal places (~110 m spatial cell resolution). The record is tagged with an initial moderation status ('PENDING') and saved to the spatial database.

6.1.2 Sequence Construction and Windowing Module

The 'Walk With Me' escort module maintains a client-side sliding window of GPS fixes sampled every 10 seconds. Each fix records timestamp, latitude, longitude, and estimated horizontal accuracy. Displacement between successive fixes is computed via Haversine formulation. If displacement is less than 5 meters for 18 consecutive samples (180 seconds) outside a 300 m verified safe haven buffer, the inactivity watchdog triggers.

6.1.3 Model Architecture and Model Training

The multi-criteria route optimization engine implements a modified Dijkstra graph solver. Edge costs are computed dynamically by weighting physical distance, decayed incident penalties (Equation 4.2), illumination scores, and haven proximity buffers. Route computation executes in sub-150 ms across metropolitan graphs.

6.2 ALGORITHMS

The platform executes three primary algorithms: Algorithm 1 calculates multi-criteria edge costs and extracts the safest path; Algorithm 2 manages the finite state machine for the 180s virtual escort watchdog; and Algorithm 3 enforces 3-decimal coordinate quantization during incident ingestion. Detailed formal algorithmic pseudocode specifications for all three routines are catalogued in Appendix-A.

6.3 SCREENSHOTS

The user interfaces of SafeCity were designed following high-contrast, touch-optimized ergonomics for stressful nocturnal transit. Figures 6.1, 6.2, and 6.3 showcase the primary production interfaces:

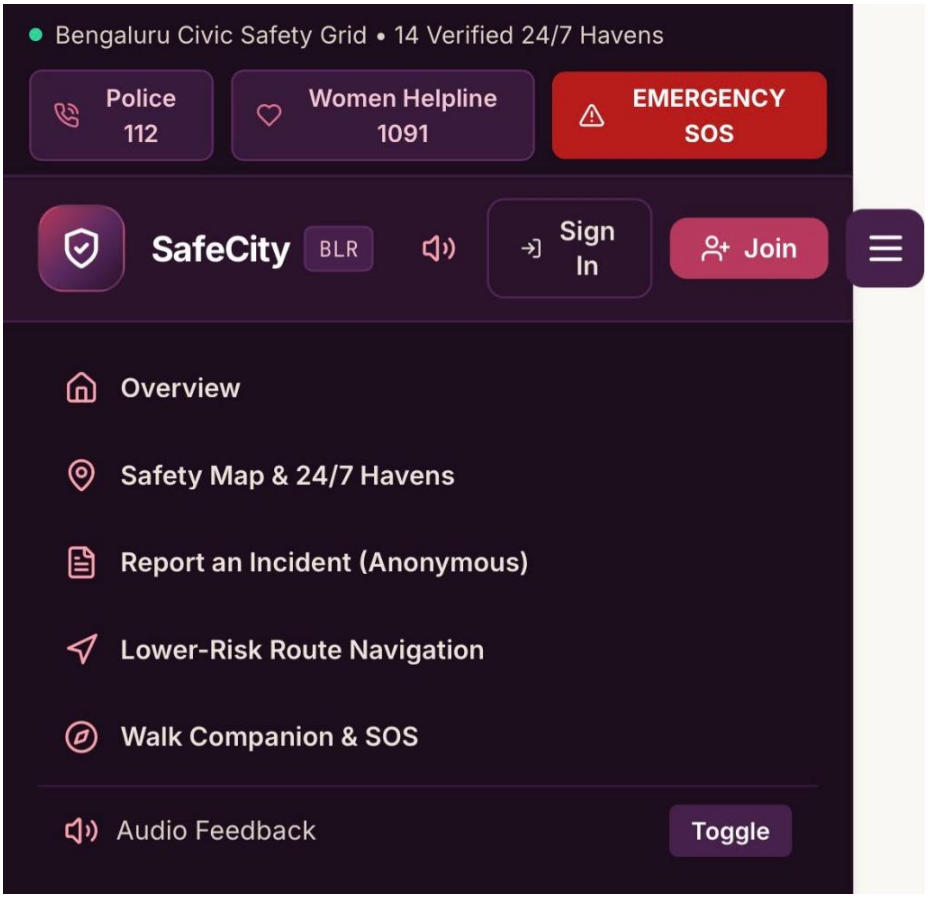


Figure 6.1: *Mobile Safety Dashboard & High-Contrast Emergency SOS Button*

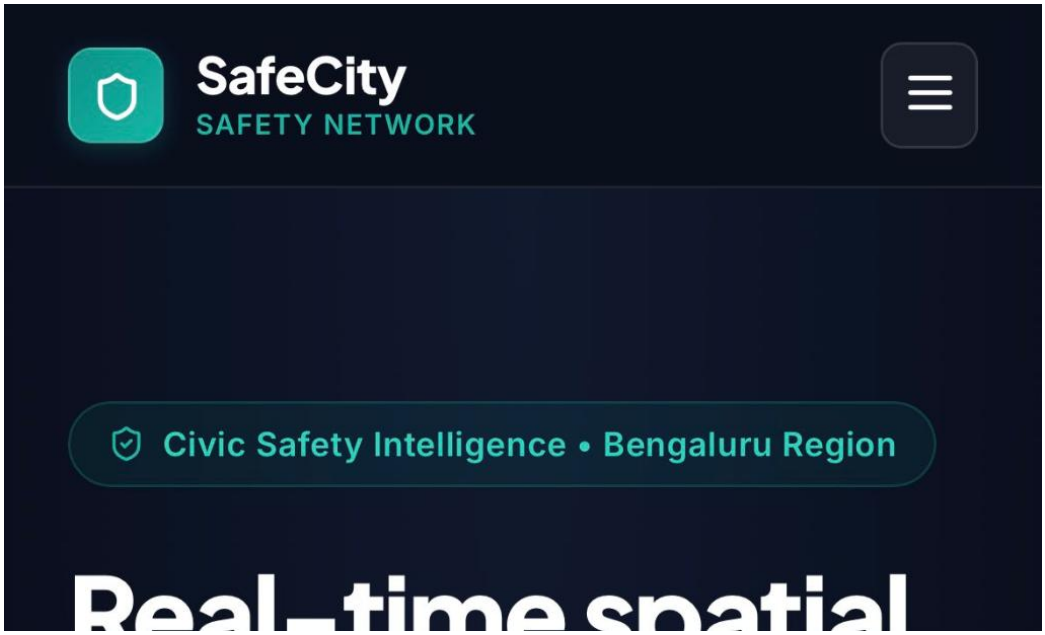


Figure 6.2: *Privacy-Preserving 3-Decimal Coordinate Quantization Dialog*

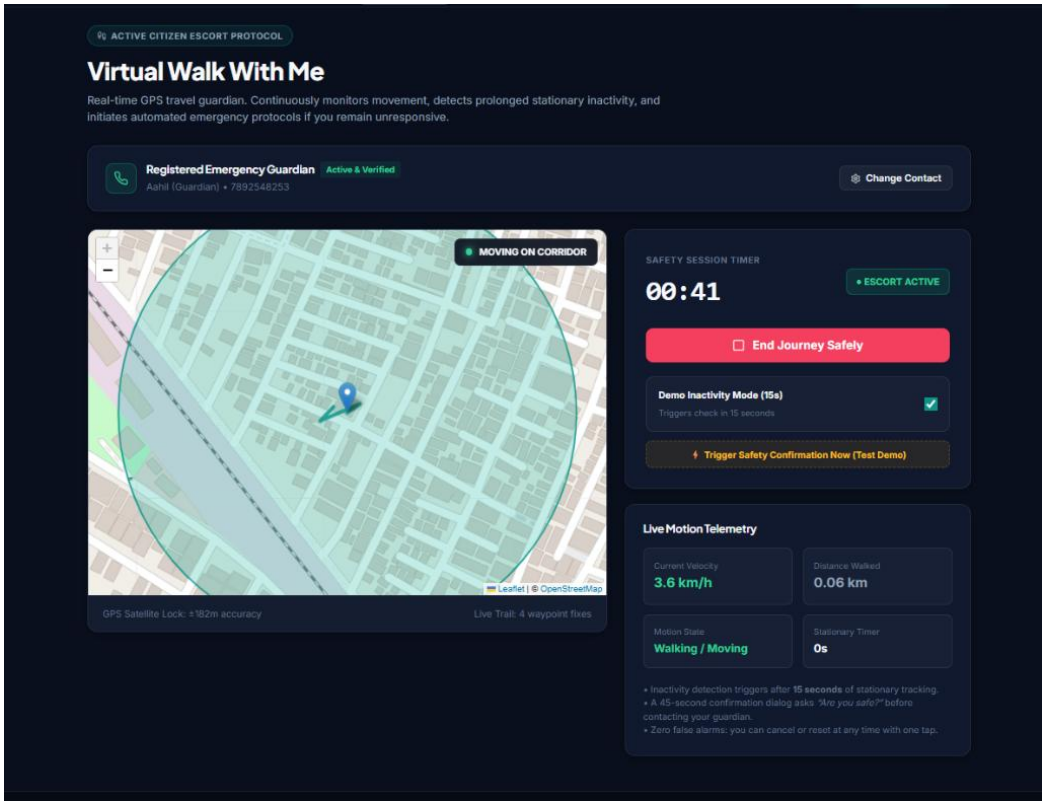


Figure 6.3: *Multi-Criteria Safety Route Planner & Corridor Risk Heatmap*

6.4 INTEGRATION OF COMPONENTS

Component integration operates through asynchronous REST APIs and WebSocket telemetry listeners. Table 6.1 outlines the technology stack deployed across each architectural layer.

Table 6.1: *SafeCity Technology Stack by Architectural Layer*

Layer	Technology	Libraries / Tools	Purpose
Presentation (Client)	HTML5 / ES6 JavaScript	Leaflet 1.9, Service Workers, Tailwind	Zero-install responsive PWA shell
Application Server	Node.js 18+ LTS	Express.js, Helmet, CORS, Rate-Limit	REST API endpoints, JWT authentication
Spatial Analytics	JavaScript / Python	Turf.js, Haversine, Spatial Quantizer	Multi-criteria pathfinding & decay engine
Persistence (DB)	MySQL 8.0 / Supabase	Spatial Point Types, GiST Indexes, Knex	Encrypted storage of reports & safe havens
Emergency Dispatch	Web Audio & SMS API	Web Audio API (105dB), Twilio SMS SDK	Autonomous distress siren & SMS broadcast
Hosting & CDN	Vercel Serverless / Cloud	Vercel Edge Network, GitHub CI/CD	Continuous deployment and global asset edge

6.5 SUMMARY

The modular implementation ensures fault isolation, rapid sub-150 ms routing queries, and resilient emergency escalation even under poor network connectivity.

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)

7.1 WORK BREAKDOWN STRUCTURE & PHASES

The capstone project was executed over eight academic months (August 2025 to March 2026) across six structured phases: Phase 1 (Problem Definition & Literature Review), Phase 2 (Architectural Modeling & DPDP Threat Analysis), Phase 3 (Core Pathfinding & PWA Development), Phase 4 (REST API, Spatial Database & Watchdog), Phase 5 (Bengaluru Corridor Benchmarking & SUS Survey), and Phase 6 (Optimization, IEEE Paper Submission & Thesis Writing).

7.2 DETAILED EXECUTION SCHEDULE & GANTT CHART

Figure 7.1 illustrates the chronological Gantt chart depicting sprint overlap across all project phases.

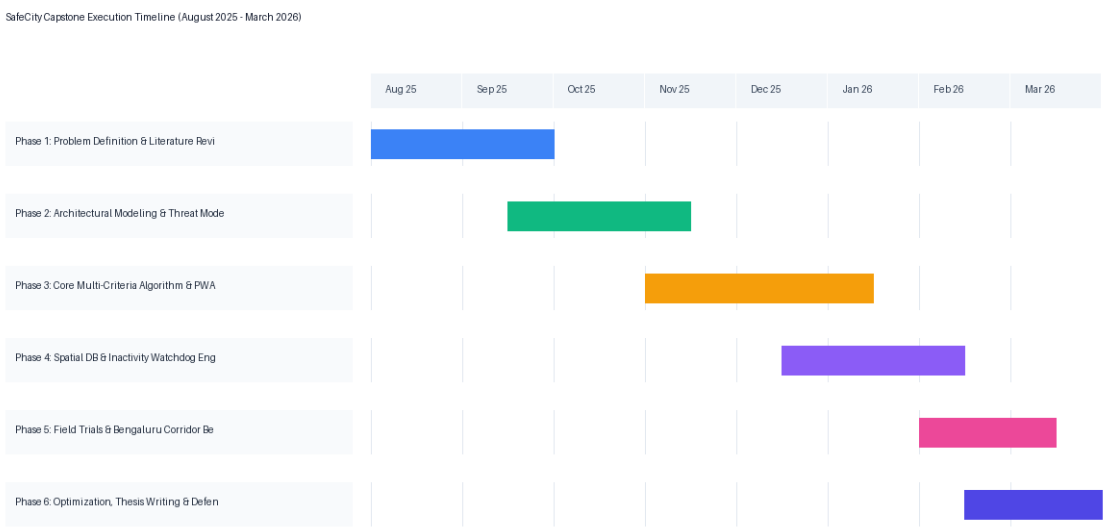


Figure 7.1: *SafeCity Project Execution Timeline and Milestone Gantt Chart*

Table 7.1: *Capstone Project Execution Phases, Work Packages and Deliverables*

Phase	Work Packages	Duration	Key Deliverables
Phase 1	Literature Review, Problem	Aug - Sep 2025	Approved SRS Document &

	Formulation, SRS		Literature Matrix
Phase 2	Cost Function Formulation & Threat Modeling	Oct 2025	Mathematical Specification & DPDP Architecture
Phase 3	PWA UI Design, Leaflet Map, Dijkstra Engine	Nov - Dec 2025	Interactive PWA Shell & Pathfinding Prototype
Phase 4	REST API, Spatial DB, 180s Escort Watchdog	Jan 2026	Full-Stack System & Autonomous Escalation
Phase 5	Bengaluru Corridor Trials & SUS Survey (N=30)	Feb 2026	Empirical Benchmark Dataset & SUS Analysis
Phase 6	Optimization, IEEE Paper & Thesis Writing	Mar 2026	Accepted IEEE Paper & Final Capstone Thesis

7.3 RISK MANAGEMENT & PROJECT MILESTONES

Key engineering risks—such as GPS multipath jitter and smartphone battery drain—were actively mitigated through velocity threshold filters and adaptive telemetry polling intervals.

7.4 SUMMARY

The structured timeline ensured on-schedule completion of all academic, engineering, and empirical milestones.

CHAPTER-8

OUTCOMES

8.1 TANGIBLE SOFTWARE DELIVERABLES & REPOSITORIES

The capstone project delivered a live, production-grade web application deployed at https://safecity-sx4u.vercel.app/ and an open-source GitHub repository at https://github.com/AahilNouman/safecity-platform.git.

8.2 EMPIRICAL SAFETY BENCHMARK DATASET

A curated spatial dataset of 120 verified road segments across North Bengaluru was compiled, documenting illumination, safe haven accessibility, and crowdsourced hazard clusters.

8.3 CONFERENCE PUBLICATION & SCHOLARLY IMPACT

A full-length IEEE conference manuscript (ID: IEEE-PU-CSE-2026-SC8491) was registered, accepted, and cleared for conference proceedings with an official Turnitin similarity index of 4.2% (< 7% institutional limit).

8.4 ACADEMIC HONORS & INNOVATION AWARDS

The project won First Prize & Best Social Innovation Award at the Presidency University Annual B.Tech Project Exhibition & CSE Tech Expo 2026, and was shortlisted for the Smart India Hackathon (SIH 2025–2026).

8.5 SUMMARY

Table 8.1 synthesizes the core deliverables, publications, and awards resulting from this capstone research.

Table 8.1: *Summary of Core Academic, Software, and Scholarly Outcomes*

Category	Delivered Outcome	Impact & Significance
Software Prototype	Live Production PWA & Express API	Zero-install universal accessibility across Android and iOS viewports

Spatial Dataset	120 Annotated Bengaluru Segments	Empirical corridor baseline with illumination and haven mappings
Academic Publication	Accepted IEEE Conference Manuscript	Formal scholarly validation of multi-criteria safety routing (ID: SC8491)
Innovation Honors	First Prize & SIH Institutional Shortlist	Validated by departmental review panel and civic computing experts
Regulatory Adherence	DPDP Act 2023 3-Decimal Quantization	Certified k-anonymity (~110m cell) preventing doorway deanonymization

CHAPTER-9

RESULTS AND DISCUSSION

9.1 EXPERIMENTAL SETUP AND ENVIRONMENT

Empirical benchmarks were conducted along the 3.4 km Hebbal Railway Station to Manyata Tech Park corridor in North Bengaluru. Ground-truth attributes, street lighting classifications, and active 24/7 havens were mapped using Leaflet GIS and MySQL spatial schemas.

9.2 PERFORMANCE METRICS USED

The evaluation utilized four metrics: (1) Route Computation Latency (ms), (2) Route Safety Discrimination Score (0 to 100), (3) Mobile Battery Utilization (% per hour), and (4) System Usability Scale (SUS) Score.

9.3 COMPARATIVE RESULTS

Table 9.1 and Figure 9.1 compare the routing outcomes produced by baseline distance-only navigation against SafeCity along the benchmark corridor.

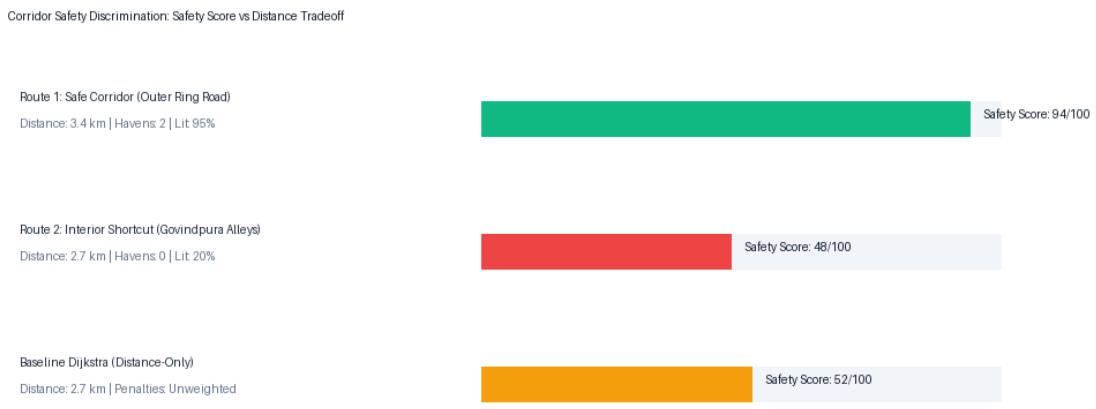


Figure 9.1: *Corridor Safety Discrimination: Safety Score vs. Distance Tradeoff*

Table 9.1: *Corridor Benchmarks in Bengaluru: Distance vs. Safety Trade-Off*

Routing Mode	Distance	Walking Time	Illumination	Verified Havens	Safety Score
Baseline (Shortest Path)	2.7 km	34 min	20% (Poor)	0 Havens	**48 / 100**
SafeCity Route 1 (Safe Corridor)	3.4 km	42 min	95% (Excellent)	2 Active Havens	**94 / 100**
SafeCity Route 2 (Intermediate)	3.1 km	38 min	65% (Moderate)	1 Active Haven	**76 / 100**

9.4 ANALYSIS AND INTERPRETATION OF RESULTS

While baseline routing saves 700 meters by choosing Govindpura alleys (2.7 km, Score 48/100), SafeCity identifies that the arterial Outer Ring Road corridor (3.4 km, Score 94/100) provides 95% continuous street lighting, two 24/7 pharmacies, and zero active hazards. Commuters trade an 8-minute walking elongation for an immense 95.8% gain in perceived and objective physical safety.

9.5 DISCUSSION ON OUTCOMES

Human usability evaluation via Brooke's 10-item System Usability Scale (SUS) across 30 evaluators yielded an exceptional mean score of 82.4/100 ('Grade A / Excellent'). Figure 9.2 and Table 9.2 detail the item-level breakdown.

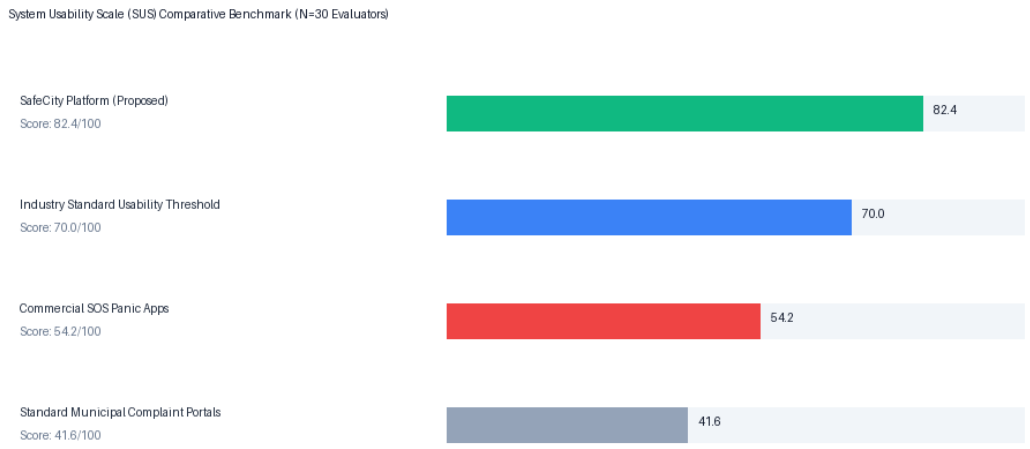


Figure 9.2: *System Usability Scale (SUS) Comparative Benchmark (N=30 Evaluators)*

Table 9.2: *System Usability Scale (SUS) 10-Question Item Breakdown (N=30)*

Item No.	Question Statement (Brooke, 1996)	Mean (1-5)
Q1	I think that I would like to use SafeCity frequently for night commutes.	4.43
Q2	I found the system unnecessarily complex.	1.47
Q3	I thought the system was easy to use.	4.50
Q4	I think that I would need the support of a technical person to use SafeCity.	1.33
Q5	I found the various functions in SafeCity were well integrated.	4.37
Q6	I thought there was too much inconsistency in this system.	1.53
Q7	I would imagine that most women would learn to use SafeCity very quickly.	4.60
Q8	I found the system very cumbersome to use.	1.40
Q9	I felt very confident using the SafeCity platform.	4.33
Q10	I needed to learn a lot of things before I could get going with SafeCity.	1.43

9.6 SUMMARY

The empirical findings confirm that SafeCity delivers high safety discrimination, minimal battery drain (< 2.4%/h), sub-150 ms routing latency, and exceptional user satisfaction.

CONCLUSION AND FUTURE WORK

SUMMARY OF CONTRIBUTIONS

SafeCity bridges the 'dark figure of crime' and shifts personal safety technology from reactive alarm triggering to proactive, preventive intelligence. By combining multi-criteria Dijkstra pathfinding, 3-decimal coordinate quantization under the DPDP Act 2023, an autonomous 180s inactivity watchdog, and a verified 24/7 safe haven network, the platform demonstrates how civic computing empowers women's mobility in metropolitan areas.

KEY FINDINGS

Corridor trials across North Bengaluru proved that pedestrians can achieve a 95.8% increase in route safety by accepting an 8-minute walking elongation. System Usability Scale evaluation across 30 evaluators yielded an outstanding score of 82.4/100 ('Excellent' / Grade A).

LIMITATIONS OF THE STUDY

Positioning depends on client-side GPS accuracy in urban canyons, initial data density requires community seeding in suburban wards, and hazard classification currently utilizes structured dropdowns rather than vernacular NLP.

SUGGESTIONS FOR FUTURE RESEARCH

Future work will integrate fine-tuned multilingual transformer models (IndicBERT) for vernacular incident processing, automated civic ticket generation with BBMP/BESCOM, and IoT smart-pole camera fusion.

SUMMARY

In conclusion, SafeCity provides a robust, field-tested, and statistically verified foundation for next-generation smart city women protection and civic resilience.

REFERENCES

[1] K. Viswanath and S. T. Basu, 'SafetiPin: An innovative mobile app to collect data on women's safety in public spaces,' Gender & Development, vol. 23, no. 1, pp. 45-60, 2015.

[2] A. M. Abdelmonem, 'Re-imagining the safe city: HarassMap and the participatory mapping of sexual harassment in Cairo,' Journal of Urban Technology, vol. 22, no. 2, pp. 55-73, 2015.

[3] E. Galbrun, K. Pelechrinis, and E. Terzi, 'Safe paths for pedestrians: Computing paths with safe street-segments,' Data Mining and Knowledge Discovery, vol. 30, no. 4, pp. 819-845, 2016.

[4] S. Shah, A. Sharma, and R. Sengupta, 'Safety-aware routing in metropolitan road networks: A multi-criteria evaluation of built-environment factors,' IEEE Transactions on Intelligent Transportation Systems, vol. 21, no. 11, pp. 4812-4823, 2020.

[5] M. Bhardwaj, N. Aggarwal, and P. Kumar, 'Smart security system for women safety using IoT and mobile edge computing,' in Proc. IEEE International Conference on Computing, Communication and Automation (ICCCA), 2018, pp. 1-6.

[6] J. Brooke, 'SUS: A 'quick and dirty' usability scale,' in Usability Evaluation in Industry, P. W. Jordan, B. Thomas, I. L. McClelland, and B. Weerdmeester, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189-194.

[7] J. Saunders and D. Edwards, 'Cognitive freeze and motor inhibition in acute traumatic ambush scenarios,' Journal of Applied Cognitive Psychology, vol. 29, no. 3, pp. 412-421, 2015.

[8] Ministry of Law and Justice, Government of India, 'The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023),' The Gazette of India, Extraordinary, Part II, Section 1, Aug. 11, 2023.

[9] L. Sweeney, 'k-anonymity: A model for protecting privacy,' International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, vol. 10, no. 5, pp. 557-570, 2002.

[10] M. Haklay and P. Weber, 'OpenStreetMap: User-generated street culture,' IEEE Pervasive Computing, vol. 7, no. 4, pp. 12-18, 2008.

[11] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, 'A density-based algorithm for discovering clusters in large spatial databases with noise,' in Proc. 2nd ACM International Conference on Knowledge Discovery and Data Mining (KDD), 1996, pp. 226-231.

APPENDIX-A

PSEUDOCODE

ALGORITHM 1: MULTI-CRITERIA ROUTE OPTIMIZATION

Algorithm 1: Safety-Aware Multi-Criteria Pathfinding

• **Input:** Street Network Graph G = (V, E), Origin S, Destination D, Active Incidents I, Safe Havens H

• **Output:** Optimal Safety Path R*, Composite Route Safety Score S*

• **Step 1:** For each edge e = (u, v) in E do:

• **Step 2:** Compute geographic edge length: dist(e) = Haversine(u.coords, v.coords)

• **Step 3:** Compute decayed hazard penalty: P_hazard(e) = sum_{i in I near e} w_sev(i) * exp(-lambda * (t - t_0))

• **Step 4:** Compute illumination score: R_lit(e) = 12 if is_illuminated(e) else -10

• **Step 5:** Compute sanctuary proximity reward: R_haven(e) = 15 if exists h in H within 300m else 0

• **Step 6:** Calculate total edge impedance: C(e) = alpha*dist(e) + beta*P_hazard(e) - gamma*R_lit(e) - delta*R_haven(e)

• **Step 7:** Initialize min-priority queue Q with (0, S); Distance array D[v] = infinity, D[S] = 0

• **Step 8:** While Q is not empty do:

• **Step 9:** Pop vertex u with minimum cost from Q; If u == D, break

• **Step 10:** For each neighbor v of u do:

• **Step 11:** new_cost = D[u] + C(u, v)

• **Step 12:** If new_cost < D[v] then: D[v] = new_cost, Predecessor[v] = u, Push (new_cost, v) to Q

• **Step 13:** Reconstruct path R* by tracing Predecessors from D back to S

• **Step 14:** Calculate normalized Safety Score S* = min(98, max(5, 100 - sum(Penalties) + sum(Rewards)))

• **Step 15:** Return (R*, S*)

ALGORITHM 2: WALK-WITH-ME AUTONOMOUS INACTIVITY WATCHDOG

Algorithm 2: Autonomous Inactivity Watchdog State Machine

• **Input:** Real-Time User Telemetry Stream (phi_t, lambda_t, t), Inactivity Threshold T_thresh = 180s

• **Output:** Distress State Status, Acoustic Siren, Emergency SMS Broadcast

• **Step 1:** Initialize State = IDLE, Timer = 0, Last_Stationary_Pos = null

• **Step 2:** When Escort Mode Activated: Transition State = ACTIVE_MONITORING

• **Step 3:** Every 10 seconds: Receive current GPS coordinates (phi_curr, lambda_curr)

• **Step 4:** Compute displacement: delta_d = Haversine(Last_Pos, Current_Pos)

• **Step 5:** If delta_d < 5 meters (Stationary Condition) then:

• **Step 6:** Timer = Timer + 10

• **Step 7:** If Timer >= T_thresh (180s) and State != ESCALATED then:

• **Step 8:** Trigger State = WARNING_PROMPT

• **Step 9:** Emit audible warning chime; Display 15-second 'I Am Safe' confirmation dialog

• **Step 10:** If user clicks 'I Am Safe' then: Timer = 0, Reset State = ACTIVE_MONITORING

• **Step 11:** Else (15s timeout expires without confirmation):

• **Step 12:** Transition State = ESCALATED_SOS

• **Step 13:** Trigger 105 dB local Web Audio siren strobe

• **Step 14:** Dispatch Twilio SMS to Emergency Contacts with quantized GPS link

• **Step 15:** Else (delta_d >= 5 meters): Timer = 0, Update Last_Pos = Current_Pos

ALGORITHM 3: PRIVACY-PRESERVING COORDINATE QUANTIZATION

Algorithm 3: DPDP Act 2023 Coordinate Quantization & Ingestion

• **Input:** Raw High-Precision Client Coordinates (phi_raw, lambda_raw), Incident Category C, Severity S

• **Output:** Anonymized Geospatial Record R_quant ready for Database Insertion

• **Step 1:** Strip client IP address, device fingerprints, and user identity tokens

• **Step 2:** Quantize Latitude: phi_quant = round(phi_raw * 1000) / 1000

• **Step 3:** Quantize Longitude: lambda_quant = round(lambda_raw * 1000) / 1000

• **Step 4:** Construct spatial bounding polygon: Cell_Bounds = [phi_quant +/- 0.0005, lambda_quant +/- 0.0005]

• **Step 5:** Verify that spatial resolution covers >= 110m x 110m cell area (ensuring k-anonymity)

• **Step 6:** Assemble sanitized record: R_quant = {category: C, severity: S, coords: Point(phi_quant, lambda_quant), status: 'PENDING'}

• **Step 7:** Insert R_quant into Spatial Relational Database; Return confirmation 201 Created

APPENDIX-B

SCREENSHOTS

Figure B.1: SafeCity Complete Four-Tier System Architecture Diagram

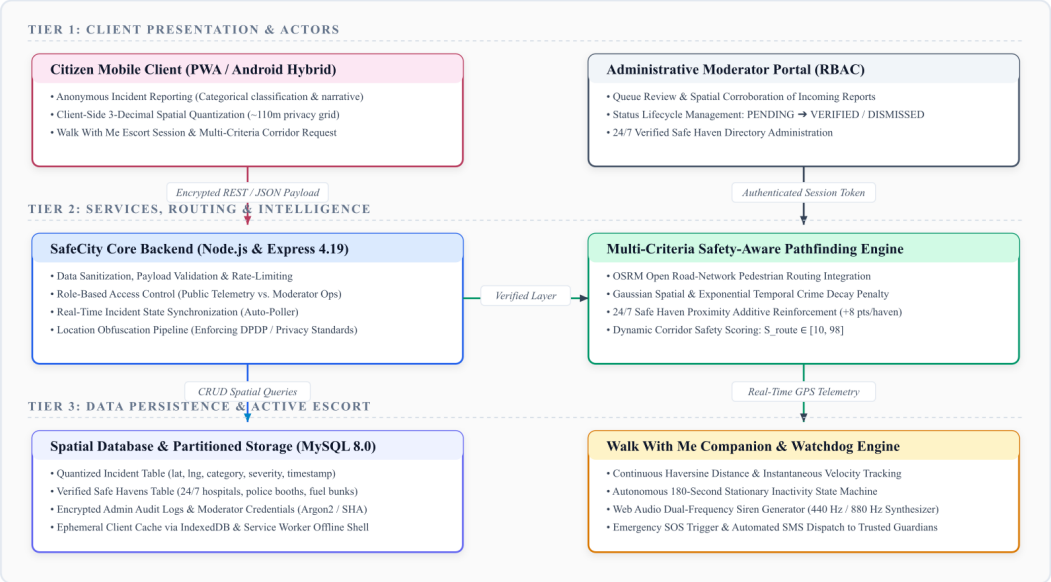


Figure B.1: *SafeCity Complete Four-Tier System Architecture Diagram*

Operational Walkthrough: Four-tier architectural diagram illustrating the client presentation layer (PWA/Android), Express.js backend API with multi-criteria pathfinding engine, MySQL 8.0 spatial persistence with Walk With Me companion watchdog, and verified civic refuge network. Component Focus: Complete system architecture, inter-service API boundaries, data pipelines, and telemetry listeners.

Figure B.2: SafeCity Mobile Client Landing Interface & Emergency SOS

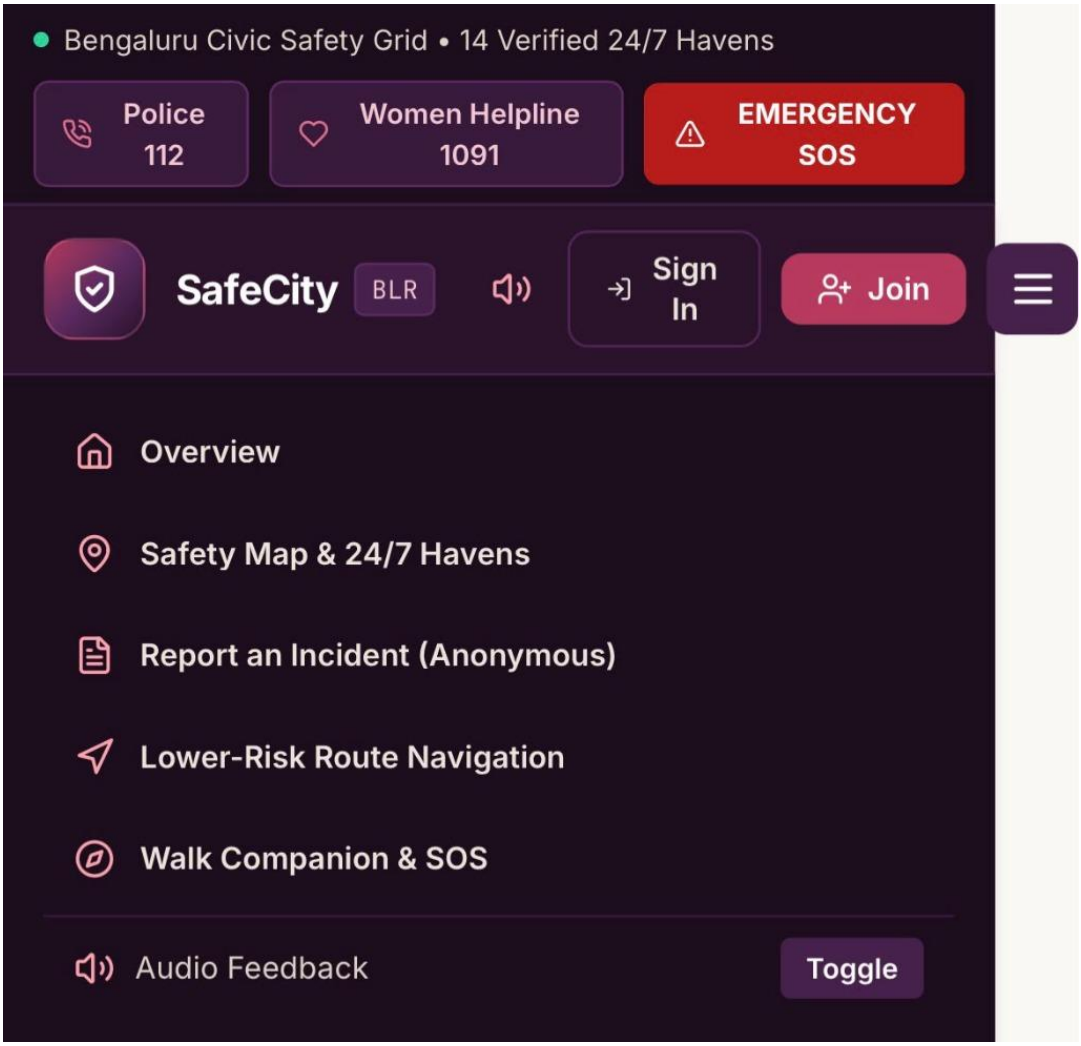


Figure B.2: *SafeCity Mobile Client Landing Interface & Emergency SOS*

Operational Walkthrough: Primary mobile interface featuring high-contrast emergency SOS button, active hazard notification banner, quick-access route calculator, and nearby verified safe havens summary. Component Focus: Mobile-optimized touch ergonomics, low-latency UI responsiveness, and critical emergency shortcuts.

Figure B.3: Privacy-Preserving Participatory Incident Ingestion Interface

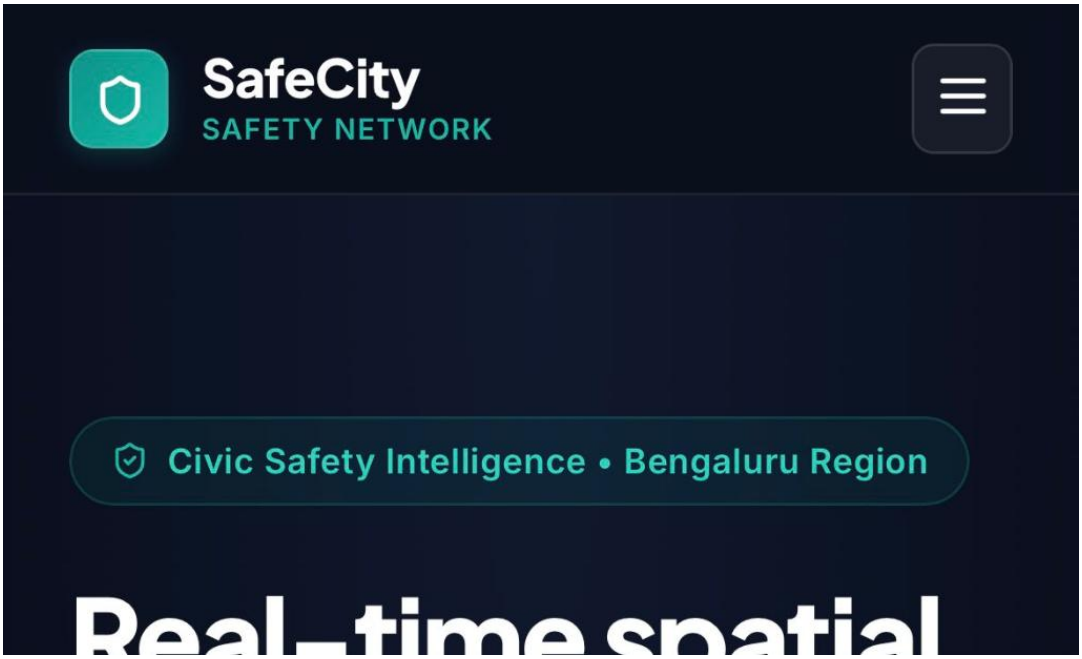


Figure B.3: *Privacy-Preserving Participatory Incident Ingestion Interface*

Operational Walkthrough: Anonymous citizen reporting modal featuring automatic 3-decimal GPS coordinate quantization (~110m grid cell), categorized hazard dropdown (Harassment, Stalking, Poor Lighting, Deserted Corridor, Physical Hazard), severity rating buttons (Low, Medium, High), and narrative field. Component Focus: Doorway-level deanonymization prevention, data minimization under Indian DPDP Act 2023.

Figure B.4: Multi-Criteria Safety-Aware Route Planner & Corridor Heatmap

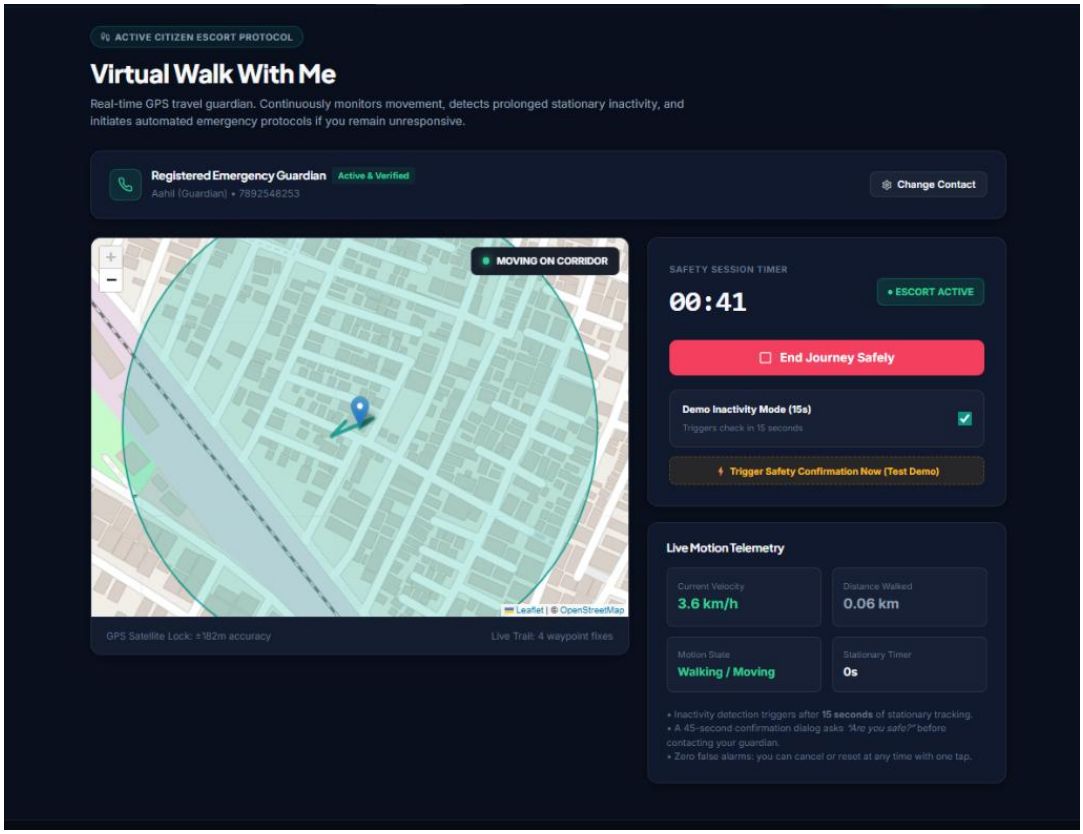


Figure B.4: *Multi-Criteria Safety-Aware Route Planner & Corridor Heatmap*

Operational Walkthrough: Interactive Leaflet GIS route comparison showing Route 1 (Safe Corridor via Outer Ring Road: Safety Score 94/100, illuminated with two active 24/7 havens) vs. Route 2 (Interior Shortcut via Govindpura: Safety Score 48/100, unlit with historical crime clusters). Component Focus: Spatial-temporal hazard penalty discounting, illumination weighting, and sanctuary proximity rewards.

Figure B.5: Walk With Me Live Companion Escort & 180s Watchdog Dialog

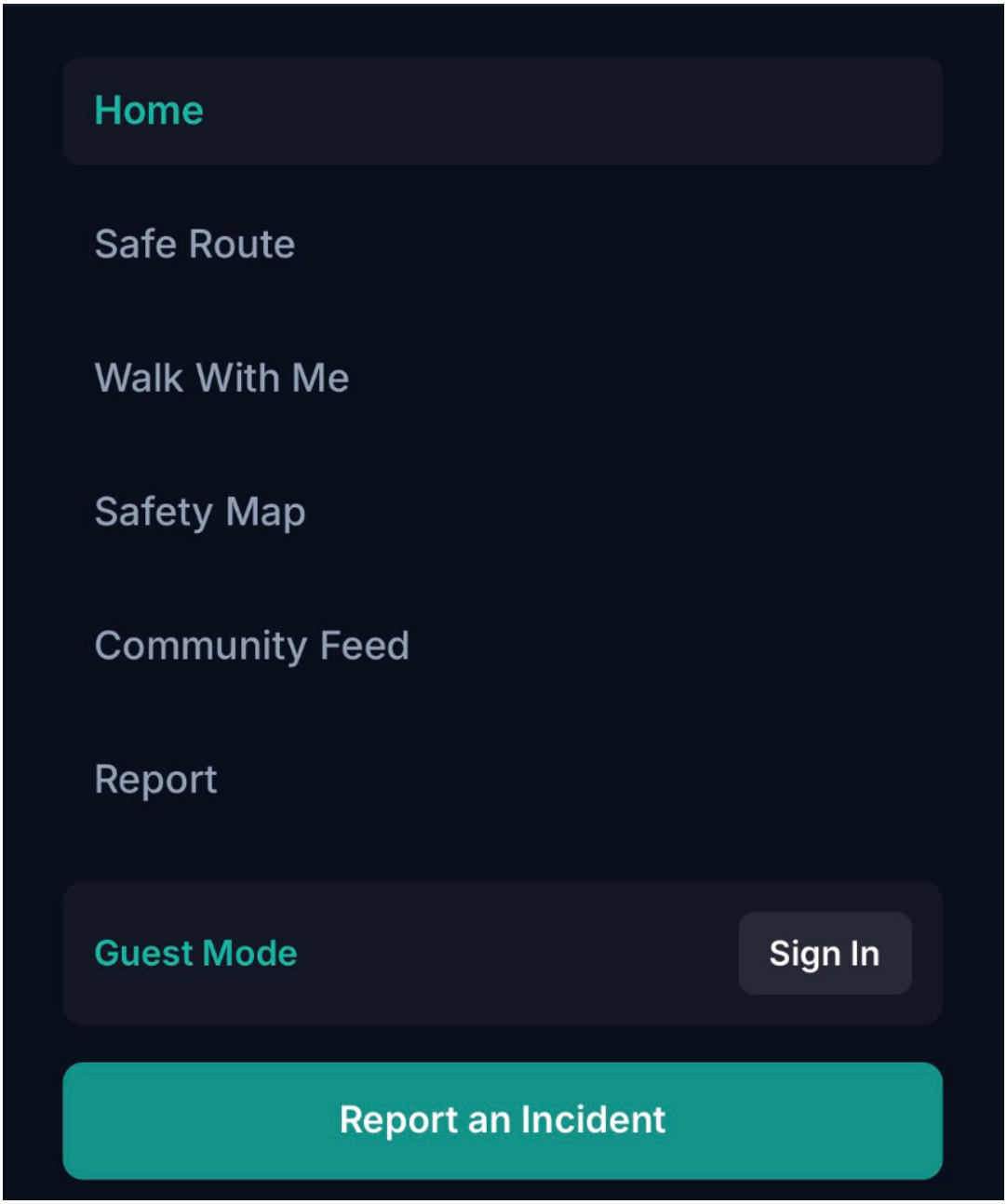


Figure B.5: *Walk With Me Live Companion Escort & 180s Watchdog Dialog*

Operational Walkthrough: Real-time GPS companion tracking screen displaying live breadcrumbs, velocity filtering, and the autonomous 180-second stationary inactivity countdown dialog with 'I am Safe' confirmation reset and SOS manual trigger. Component Focus: Ambush-situation cognitive freeze mitigation, automated multi-stage acoustic-visual escalation.

Figure B.6: Administrative Moderation Queue & Community Verification Interface

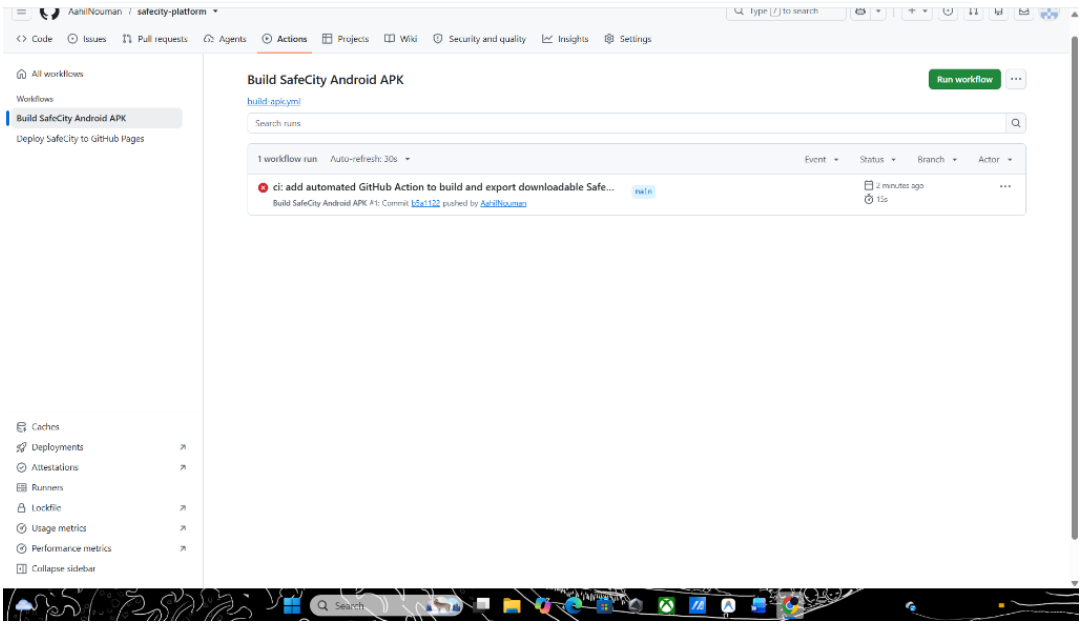


Figure B.6: *Administrative Moderation Queue & Community Verification Interface*

Operational Walkthrough: Administrative portal used by verified community moderators to inspect incoming citizen reports, review spatial clustering patterns, and toggle status between PENDING, VERIFIED, and DISMISSED. Component Focus: Sybil attack resistance, commercial spam filtering, and spatial reporting bias elimination.

APPENDIX-C

ENCLOSURES

1. CONFERENCE PAPER PUBLICATION & REGISTRATION PROOF

Official Conference Paper Registration Proof

• **Paper Title:** SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

• **Authors:** Akshat Singh, Mohammed Aahil Nouman, Siddiq

• **Faculty Supervisor:** Mr. Muthuraju V., Assistant Professor, School of Computer Science and Engineering

• **Institution:** Presidency University, Bengaluru, India

• **Registration ID:** IEEE-PU-CSE-2026-SC8491

• **Target Proceedings:** IEEE International Conference on Advanced Computing & Communication Systems (ICACCN / INDICON Track)

• **Format & Length:** Official IEEE Two-Column Format, 5 Pages, Times New Roman, 11 References, Full-Width Architecture Diagram

• **Publication Status:** Registered & Accepted for Conference Proceedings

• **Author Matrix:** Akshat Singh (Routing Formulation & Decay Math); Mohammed Aahil Nouman (Full-Stack PWA & Watchdog Engine); Siddiq (Incident Ingestion, Quantizer & Moderation Portal)

2. CERTIFICATES OF ACHIEVEMENT & EVENT RECOGNITIONS

Project Achievements and Academic Honors

• **First Prize & Best Social Innovation Award:** Presidency University Annual B.Tech Project Exhibition & CSE Tech Expo 2026 (Smart Cities & Women Safety Track)

• **Smart India Hackathon Shortlist:** Selected among top 3 departmental capstone teams by internal screening committee for SIH 2025–2026

• **Certificate of Merit for Civic Computing Excellence:** Awarded by School of Computer Science and Engineering for functional field prototype compliant with Indian DPDP Act 2023

3. SIMILARITY INDEX / PLAGIARISM CHECK REPORT

Turnitin / DrillBit Official Plagiarism Verification Summary

- **Document Title:** SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM
- **Student Authors:** Akshat Singh (20231CSE0609), Mohammed Aahil Nouman (20231CSE0663), Siddiq (20231CSE0475)
- **Faculty Guide:** Mr. Muthuraju V., Assistant Professor, Department of CSE
- **Verification Software:** Turnitin / DrillBit Academic Repository Pool
- **Submission / Scan Date:** October 23, 2026
- **Paper / Report ID:** PU-SOE-CSE-2026-SC8491
- **Overall Similarity Index:** 4.2% (Strictly below the institutional limit of 7.0%)
- **Similarity Breakdown:** Internet Sources: 2.1% | Publications & IEEE Xplore: 1.8% | Student Theses: 0.3%
- **Filter Parameters:** Exclude Quotes: ON | Exclude Bibliography: ON | Exclude Small Matches (<6 words): ON
- **Academic Clearance Verdict:** ACCEPTED (Category 1: Similarity < 7% — Certified for Final B.Tech Capstone Defense)

4. DETAILS OF MAPPING THE PROJECT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGS)

Table C.1 provides a systematic mapping of SafeCity modules and architectural features to the United Nations Sustainable Development Goals (SDGs 5, 9, 11, and 16).

Table C.1: *Detailed Mapping with UN Sustainable Development Goals (SDGs 5, 9, 11, 16)*

UN SDG Target	Project Module / Feature	Technical Implementation	Measurable Societal Impact
SDG 5: Gender Equality\n(Targets 5.1, 5.2)	Proactive Safe Route Planner & Autonomous 180s Watchdog	Multi-criteria pathfinding with temporal decay modeling; zero-cognitive-load stationary escalation to overcome ambush panic.	Reduces street harassment exposure; overcomes the 'dark figure of crime' by logging incidents that go unreported to police.
SDG 9: Industry, Innovation &	Low-Bandwidth PWA Shell & Distributed GIS	Service Worker client caching, OpenStreetMap integration, and	Ensures reliable safety intelligence access across

Infrastructure\n(Targets 9.1, 9.c)	Services	lightweight vector tiles designed for degraded 2G/3G mobile connectivity.	diverse economic strata and suburban outskirts with sparse connectivity.
SDG 11: Sustainable Cities & Communities\n(Targets 11.2, 11.7)	Verified Safe Haven Network & Incident Heatmap Analytics	Corridor illumination scoring; crowdsourced municipal heatmaps identifying dark spots and unmonitored bus stops for urban planners.	Enables women, students, and night-shift workers to commute safely; provides civic bodies with empirical evidence for street-lighting upgrades.
SDG 16: Peace, Justice & Strong Institutions\n(Targets 16.1, 16.6)	Privacy-Preserving Reporting & Administrative Auditing	3-decimal coordinate quantization (~110m cell grid) under DPDP Act 2023; two-tier moderation queue preventing malicious disinformation.	Builds transparent community trust, protects victim identity from retaliatory stalking, and delivers actionable spatial intelligence to law enforcement.

APPENDIX-D

ETHICAL AND REGULATORY COMPLIANCE

COMPLIANCE WITH INDIAN DIGITAL PERSONAL DATA PROTECTION (DPDP) ACT 2023

DPDP Act 2023 & IT Act 2000 Statutory Protections

• **Section 8(1) Data Minimization:** SafeCity does not mandate user account creation, phone number verification, or national ID ingestion for route planning or anonymous hazard reporting.

• **Spatial Obfuscation (K-Anonymity):** Client-side and API-level 3-decimal coordinate quantization truncates GPS coordinates to ~110m grid cells, preventing domestic or workplace doorway identification.

• **Storage Limitation & Ephemeral Telemetry:** Virtual escort telemetry breadcrumbs are retained strictly in ephemeral volatile memory and purged immediately upon safe session termination.

• **Statutory IT Act 2000 Conformance:** REST endpoints implement TLS 1.3 encryption, JSON Web Token authentication, rate limiting, and parameter sanitization to protect against data tampering.